

4

reading the data

What's the Story?

out of Power Query, into Excel, find the answer

Three days getting the data in shape. Today you finally get to find out what it says. The cleaned, merged tables you built yesterday are about to leave Power Query and land in Excel — where formulas, pivots and charts live.

Today's job is *descriptive statistics*. Not the scary kind. The useful kind — counts, totals, averages, medians, mins, maxes. The numbers Priya is going to ask you for at the end of the week. Plus the ones she *didn't* ask for, but that you spot in the data and bring to her anyway. That last bit is what makes a good data tech.

We'll meet a small family of Excel functions, learn when each one is the right tool, and bump into the most common stats trap: **when the mean lies**. By the end of today, the year-on-year story Priya wants will be written in numbers — ready for Friday afternoon's charts.

Wait — we're leaving Power Query?

— you, Thursday morning, slightly attached to PQ now

Only for analysis. PQ has done its job — now Excel takes the baton.

the right tool for the right job.



BRAIN POWER

Priya is going to want three numbers from you on Friday: a churn rate, a watch-time figure, and something about which titles are most-watched. Before you turn the page, jot down which Excel function you'd reach for first to answer each one.

turn over — and we step out of Power Query →

Out of the workshop

Three days of pipeline work. Today the data leaves Power Query and lands on a worksheet — where you can hover, filter, sort, and write formulas. Same data, different room.

FOR FRIDAY

- churn rate %
- total watch hours
- top-watched titles
- year-on-year Δ

— Priya wants by Fri.

Priya:

"Right — what's the data actually telling us?"

Excel — viewing_activity_all

fx =AVERAGEIF(B:B,"Premium",E:E)

subscriber	plan	title	date	duration
00012	Premium	T0033	15/09/24	63
00088	Basic	T0042	02/06/25	41
00047	Premium	T0026	19/01/24	47
00188	Standard	T0070	11/08/25	28

Average for Premium: 54.3 minutes
10,638 rows • ready for analysis

DAY 4

Close & Load

Power Query's job, once you've cleaned and merged, is to produce *tables Excel can work with*. The hand-off is one button: **Close & Load**, top-left of the Home ribbon.

- **Close & Load**. Default. Loads the current query to a new sheet as a Table.
- **Close & Load To...** Choose target: Table, PivotTable Report, PivotChart, or Connection Only. Use this when you want a PivotTable straight away (Day 5) or to keep an intermediate query unloaded.
- **Refresh**. Every time the source data changes, hit Refresh. The whole pipeline re-runs and your sheet updates. That's the value of building the pipeline upstream.

[SCREENSHOT 1 — CLOSE & LOAD TO DIALOG]

The Import Data dialog open, showing the four output options (Table / PivotTable Report / PivotChart / Only Create Connection) and the worksheet target picker underneath.

What to load today

You have several queries from Day 3. Today's analysis needs two of them on actual sheets:

- `viewing_activity_all_enriched` — the big fact table (10,638 rows). Load as a Table on a sheet called `Activity`.
- `sw_subscribers_enriched` — every subscriber with `is_churned` and cancellation context. Load as a Table on a sheet called `Subscribers`.
- Leave the rest as Connection Only. You only need worksheet copies of the two analysis tables.

SHOW THIS OFF

As you click Close & Load, narrate: *"I'm loading only the two enriched tables I need for analysis. The intermediate queries stay as Connection Only — they're stages in the pipeline, not output."* That single sentence shows you understand the architecture, not just the buttons.

Validate then Verify

Before the analysis starts, name the two checks that sit either side of it. They sound similar; they're not. Both score marks; missing either one loses them.

Validate the data — *before* the analysis

Validation asks: **is the input fit for purpose?** Are the types right? Are there nulls where there shouldn't be? Do the row counts add up? Are the values within sensible ranges? Most of this you've already done — the *Cleaning Sequence* (Day 2) is validation, the *row-count check* on every merge (Day 3) is validation. The column profile bar in Power Query is the validation dashboard. By the time the data lands in Excel, validation should be done.

Verify the answer — *after* the analysis

Verification asks: **did my method produce the right output?** Computing the answer once isn't enough — every headline number gets a sanity check from a second angle. The named technique is **The Two-Route Check**: compute the same finding two different ways and see if the numbers agree. If they do, you've verified the method works. If they don't, something is wrong with one of the routes — and finding which is its own diagnostic.

A worked Two-Route Check

The churn rate is the headline number for Priya. Route 1: `=COUNTIF(tblSubs[status], "cancelled") / COUNTA(tblSubs[subscriber_id])`. Route 2: `=COUNTA(tblCancellations[subscriber_id]) / COUNTA(tblSubs[subscriber_id])`. One counts subscribers flagged "cancelled" in the main table; the other counts rows in the cancellations table. If the merge and the de-duplication worked, both return $87/280 = 31.07\%$. If they disagree by even one, you have a row that's in one table but not the other — almost certainly a key-format problem that survived Day 2.

SHOW THIS OFF

Say the words out loud as you work. *"That's my Validate-then-Verify step — I've already validated the inputs in Power Query, and now I'm verifying the churn rate by computing it two ways."* Naming the discipline is half the credit; the actual checks are the other half. Both halves are required.

The descriptive five

Five Excel functions answer 80% of the questions you'll be asked about a dataset. They live in the Formulas ribbon under *More Functions* → *Statistical*, or you can just type them.

Function	Answers the question	StreamWave example
<code>=COUNT (range)</code>	How many?	Total viewing sessions
<code>=SUM (range)</code>	Total?	Total minutes watched across the year
<code>=AVERAGE (range)</code>	Typical?	Mean session length
<code>=MIN (range) / =MAX (range)</code>	Smallest? Largest?	Shortest and longest sessions
<code>=COUNTA (range)</code>	How many non-empty?	Sessions where a device was recorded

A note on ranges

When your data is in a proper Excel Table (with a name like `tblActivity`), reference columns by name in formulas: `=AVERAGE(tblActivity[duration])`. Cleaner than `=AVERAGE(E2:E10638)`, and the formula automatically extends when the table grows on Refresh.

SHOW THIS OFF

Whenever you write a formula, prefer the table-column reference: "I'm using `tblActivity[duration]` rather than `E2:E10638`, so this formula keeps working when next month's data lands."

Conditional stats

Each of the descriptive five has a sister function that adds a condition. Same answer, but only for rows that match a criterion. **This is how most real questions actually get answered.**

Function	Pattern	Example
COUNTIF	(range, criterion)	<code>=COUNTIF(tblSubs[status], "cancelled")</code>
SUMIF	(criterion range, criterion, sum range)	<code>=SUMIF(tblActivity[plan], "Premium", tblActivity[duration])</code>
AVERAGEIF	(criterion range, criterion, average range)	<code>=AVERAGEIF(tblContent[genre], "Drama", tblContent[avg_user_rating])</code>
COUNTIFS & siblings	multiple criteria, all must match	Active Premium subscribers in Wales

Criterion syntax: the criterion goes in quotes whenever it contains an operator. `">60"` for "more than 60".
`">="&DATE(2025,1,1)` for "on or after 1 Jan 2025" — the ampersand joins the operator and the value.

ASSESSOR FAVOURITE

The question *"What proportion of subscribers churned?"* is two formulas, not one.

`=COUNTIF(tblSubs[status], "cancelled") / COUNTA(tblSubs[subscriber_id])`. Format the result as a percentage. Be ready to explain why you used COUNTA (counts non-empty IDs = total subscribers) rather than just hard-coding 280.

When the mean lies

AVERAGE is the function people reach for first. Often it's right. Sometimes it isn't — and the failure mode is invisible if you don't think about it.

The three M's

Three different ways to describe "what's typical":

- **Mean** (=`AVERAGE`) — add them all up, divide by how many. Useful when values are spread evenly around the centre.
- **Median** (=`MEDIAN`) — the middle value when you line them all up. Useful when a few extreme values would skew the mean.
- **Mode** (=`MODE.SNGL`) — the most common single value. Useful for categories or repeated values; less useful for continuous numbers.

A worked example

Consider `watch_duration_mins`. Most sessions are 20–60 minutes. But ten subscribers left a session open overnight (~480 minutes each). The mean will be dragged upwards by those ten — telling Priya the average session is, say, 38 minutes, when really most people watch for 35. The median will give you the honest "typical" answer.



THE RULE OF THUMB

Compute **both** the mean and the median. If they're close — use mean (everyone understands it). If they're meaningfully apart — use median and *say why*. The gap between them is itself a finding.

ASSESSOR FAVOURITE

Asked "why median, not mean?", the one-sentence answer: *"Because a small number of very long sessions would pull the mean upwards and give Priya a misleading 'typical'. Median ignores the outliers and tells the true middle."*

Spread and outliers

One number to describe "typical" isn't enough. You also want to know how spread out the data is — and what the outliers look like.

Range — the simplest spread

`=MAX(range) - MIN(range)`. Tells you the full span. Sessions on StreamWave run from a minimum of, say, 8 minutes to a maximum of 480. Range = 472. Not directly useful as a single number, but very useful for spotting that the maximum is wildly different from typical — which means there are outliers worth investigating.

Standard deviation — the proper spread

`=STDEV.S(range)`. Tells you, on average, how far values stray from the mean. Small standard deviation = tightly clustered values. Large standard deviation = wide variation. For our session durations, a STDEV near the mean itself tells you the data is highly variable.

- `=STDEV.S(range)` — sample standard deviation (use this when your data is a sample of a bigger population)
- `=STDEV.P(range)` — population standard deviation (use only when you have the entire population)
- For StreamWave subscriber data, use `STDEV.S` — these are a sample of all possible viewing behaviour.

Spotting outliers

An outlier is a value far from the rest. A common rough rule: anything more than 2 standard deviations away from the mean is worth a second look. To find them in a Table column:

`=AVERAGE(tblActivity[duration]) + 2 * STDEV.S(tblActivity[duration])` gives you the rough upper outlier threshold. Use a filter on the duration column to see which rows exceed it.

SHOW THIS OFF

After computing MIN, MAX and STDEV, point at any outliers you spot: *"Eleven sessions are over 480 minutes. That's three standard deviations above the mean — either someone left their device on overnight, or we've got bad data. Worth flagging to Priya rather than letting it skew the averages."*

Top-N and Bottom-N

Almost every report ends up wanting "top 10 X" or "bottom 5 Y". Two functions and one sort technique cover this completely.

LARGE and SMALL

`=LARGE(range, n)` returns the n^{th} -largest value. `=SMALL(range, n)` returns the n^{th} -smallest. So `LARGE(range, 1)` is the same as `MAX`; `LARGE(range, 10)` is the tenth-largest value.

Useful when you want a threshold: "What's the duration of the 10th-longest session?" —

`=LARGE(tblActivity[duration], 10)`.

For top-N rows (not just values), use sort + filter

If you want the actual rows (which subscriber, which title) — not just the threshold value — sort the table descending by the column, then filter the top N visually. Or, in a separate summary table:

- Build a column of distinct subscriber IDs.
- Next to each, `=SUMIF(tblActivity[subscriber], A2, tblActivity[duration])` to sum their total watch time.
- Sort that summary table descending by total minutes. The top 10 rows are your most-engaged subscribers.

This pattern — *build a summary, then sort* — is how 90% of "top something" questions get answered without a PivotTable. Day 5 will show you the PivotTable shortcut.

ASSESSOR FAVOURITE

The question "who are your most engaged subscribers?" is one of the things Priya will want a quick answer to on Friday. Be ready with the SUMIF-then-sort approach as your back-pocket method, even if you build the dashboard version with a PivotTable.

Halfway House

Four scenarios. Pick the right function. Answers below the line; wrong answer = the page you should re-read.

1. Priya wants to know how many of the 280 subscribers cancelled at any point. Which function?

☐ **COUNT** ☐ **COUNTA** ☐ **COUNTIF** ☐ **SUM**

2. She also wants the total minutes watched by Premium subscribers only. Which function?

☐ **AVERAGE** ☐ **SUM** ☐ **SUMIF** ☐ **AVERAGEIF**

3. Watch durations are mostly 20–60 minutes but a few are 8+ hours. You want to report "typical" session length. Which function?

☐ **AVERAGE** ☐ **MEDIAN** ☐ **MODE.SNGL** ☐ **MAX**

4. You want the duration of the 5th-longest session in the whole dataset.

☐ **MAX** ☐ **LARGE** ☐ **MEDIAN** ☐ **AVERAGEIF**

Answers — check yourself

1. **COUNTIF**. `=COUNTIF(tblSubs[status], "cancelled")`. Returns the count of rows matching the criterion.

2. **SUMIF**. `=SUMIF(tblActivity[plan], "Premium", tblActivity[duration])`. Sums the duration column only for rows where plan equals Premium.

3. **MEDIAN**. The few very-long sessions would pull AVERAGE upward and mislead. MEDIAN gives the honest middle.

4. **LARGE**. `=LARGE(tblActivity[duration], 5)`. Returns the n^{th} -largest value.

Year-on-year — the payoff

This is what Monday's effort has been leading to. Three financial years of viewing activity, sitting in one combined table. A handful of formulas turn it into the story Priya wants.

Total watch hours by FY

In a small summary table beside your data, list each FY label. Beside each, a SUMIF on the YEAR or a date range. For FY 2025-26, for example:

```
=SUMIFS(tblActivity[duration], tblActivity[date], ">="&DATE(2025,4,1), tblActivity[date], "<"&DATE(2026,4,1)) / 60
```

That returns total hours watched in that FY. Repeat for FY 24-25 and FY 26-27 (partial). The three numbers are your year-on-year totals. (Note: the upper bound uses `<` against `DATE(2026,4,1)` rather than `<=` against `DATE(2026,3,31)` — this catches any sessions with a time component on 31 March that would otherwise be silently dropped. Defensive habit; assessors notice.)

Active subscribers by FY

For each FY, count distinct subscribers who watched anything: `=COUNTA(UNIQUE(...))` if you have a newer Excel, or use a helper column flagging activity in that FY and then COUNTIF.

The year-on-year growth rate

Once you have totals per FY, growth is one formula: `=(this_year - last_year) / last_year`. Format as a percentage. A positive number = StreamWave is growing. A negative number = something to worry about.

SHOW THIS OFF

When you show Priya the year-on-year number, narrate the calculation: *"Total hours in FY 25-26 was X. In FY 24-25 (Q3 sample only) it was Y. So growth was Z%, with the caveat that the prior-year sample is partial — the apples-to-apples comparison comes next year."* Naming a caveat is what makes the analysis trustworthy.



SHARPEN YOUR PENCIL write the formula — no Excel needed

For each question, write the formula you'd type. Use Table-column references (e.g. `tblSubs[region]`).

1. How many subscribers live in Wales?

2. What's the average rating of Drama titles in the content library?

3. How many viewing sessions were longer than 60 minutes?

4. What's the 5th-shortest session in the whole table?

Answers — check yourself

1. `=COUNTIF(tblSubs[region], "Wales")`
2. `=AVERAGEIF(tblContent[genre], "Drama", tblContent[avg_user_rating])`
3. `=COUNTIF(tblActivity[duration], ">60")`
4. `=SMALL(tblActivity[duration], 5)`

There are no *Dumb* Questions

Q: Why won't my AVERAGEIF accept the criterion `>100` without quotes?

A: Criteria strings in COUNTIF/SUMIF/AVERAGEIF always need quotes when they contain an operator. `">100"` works; `>100` alone is a syntax error. If you're referring to a cell value, use `">"&A1` — the ampersand concatenates the operator and the value.

Q: What's the difference between COUNT and COUNTA?

A: COUNT only counts cells with numbers. COUNTA counts cells with anything — text, numbers, dates, formulas returning text. To count subscribers you want COUNTA on the subscriber_id column (it's text). To count durations you'd use COUNT (they're numbers).

Q: If I refresh the Power Query connection, do my formulas re-evaluate?

A: Yes. When you hit Refresh All, Excel: (1) re-runs the Power Query pipeline, (2) updates the loaded Tables, (3) recalculates every formula that depends on them. This is exactly the value of doing the upstream work in Power Query — your downstream analysis stays in sync.

Q: Should I write my formulas in the Table itself or in a separate summary area?

A: Separate summary area, almost always. Adding formula columns inside the Activity Table makes the Table heavier and risks the formula not extending to new rows on refresh. Keep the Tables clean (just data) and put your derived metrics on a Summary sheet beside them.

Q: I'm getting #DIV/0! errors when computing a percentage. Why?

A: Because the denominator is zero. Wrap the formula in IFERROR: `=IFERROR(numerator/denominator, 0)` returns 0 instead of an error when the bottom is zero. Use 0 or a dash `" - "` depending on what's clearer in your report.

Q: Why are my date criteria not matching anything?

A: Almost always because the date column hasn't been recognised as a Date type. Hop back into Power Query, set the column type using Locale → English (UK), and Refresh. Day 2 was where we did this properly; if you skipped a column, fix it upstream rather than working around it downstream.

When in doubt — both the mean and the median. The gap between them tells you something the numbers alone never would.

Say it out loud

Today is the day the assessor stops asking *how* you did things and starts asking *what the data says*. Practise these three out loud.

Q1 — Walk me through what the data is telling you.

Pick two or three findings — not a long list. A churn rate. A typical session length. The most-watched genre. For each, say the number, then say what it means in plain English. *"Around 31% of subscribers have cancelled at some point — we'll display this as 31.2% once we apply number formatting on Day 7. Typical session is 38 minutes — median, not mean, because a few all-night sessions skewed the mean. Drama is the most-watched genre by total hours."*

Q2 — Why median, not mean?

One sentence. *"Because a small number of very long sessions would pull the mean upward and give a misleading 'typical'. Median ignores the outliers and tells the true middle."* Always finish with what you did about it: *"I'm reporting median for typical session length and noting the outlier count separately."*

Q3 — How would you verify your churn rate is correct?

The shape: cross-check from a different angle — the named device for today is **The Two-Route Check**. *"I calculated it two ways — once with COUNTIF on the status column, once by counting cancellation records in sw_cancellations. Both gave the same number, 87 out of 280. If they hadn't matched, I'd have known something was wrong with the merge."* Two routes to one answer is the verification that wins distinction.

SHOW THIS OFF

Use the exact phrase *"I used the Two-Route Check"* at least once during the post-demo Q&A. *"I used the Two-Route Check on the churn rate — COUNTIF on subscribers gave 87, counting cancellation rows gave 87. Same number, two routes. That's how I knew the merge was clean."* Assessors recognise verification language and credit it.



DISTINCTION TELL

For every number you quote, be ready to say *three things*: what it is, how you got it, and what it means in plain language. "*The number is 31%. I got it by dividing COUNTIF(status, 'cancelled') by the total subscriber count. It means roughly one in three subscribers has churned since launch.*" Three sentences. Always.

Bullet Points — the chapter in 30 seconds

- **Close & Load is the hand-off.** Two enriched tables to sheets, intermediate queries stay Connection Only.
- **Validate then Verify.** Validation = is the input fit for purpose (done in Power Query). Verification = did my method produce the right output (The Two-Route Check).
- **The descriptive five.** COUNT, SUM, AVERAGE, MIN, MAX answer 80% of questions.
- **The conditional five.** COUNTIF, SUMIF, AVERAGEIF (and IFS variants) answer the other 20%.
- **Mean lies on skewed data.** Always compute both mean and median; the gap is itself a finding.
- **Table-column references over cell ranges.** `tblActivity[duration]` beats `E2:E10638`.
- **Two-Route Check on every headline number.** Same answer from two different routes is verification that wins distinction.

TEST DRIVE

Before closing the laptop, run this self-check. Tick each box only when it's true:

- ☐ Two enriched queries loaded to Tables (Activity, Subscribers); the rest stay Connection Only.
- ☐ Total subscriber count and churn rate computed and **Two-Route checked** from a second angle.
- ☐ Median session length computed and compared to the mean — gap noted.
- ☐ Year-on-year hours-watched totals computed for at least two FYs.
- ☐ For every formula you wrote, you can say in one sentence what it computes and why.

COMING UP — DAY 5

Patterns & Relationships. Tomorrow we move from describing the data to *explaining* it. Correlation versus causation. Linear regression — slope, intercept, R^2 . *The Three Numbers of a Relationship*. Reading scatter plots properly. The kind of analysis that turns "here's what happened" into "here's what predicts what" — which is the level the assessor wants to see.

that's Day 4 done — the data has started talking back.